

Contact

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Top Skills

Scala
Akka
PySpark

Languages

Telugu (Limited Working)
Hindi (Limited Working)
Tamil (Limited Working)
English (Professional Working)

Certifications

Python Programming For
EveryBody

Honors-Awards

Secured 3rd position in Innovation
Entrepreneurship development
center Summit 2016
1st place in Paper Presentation at
Madras Institute of Technology

HemanthKumar K Ramakrishnan

Lead Software Engineer at Freddie Mac | Develop & Deploy Agentic
Systems | Java | Python | LLMs | RAG | MCP | LangGraph | React |
No-SQL | Spark | AWS

Aldie, Virginia, United States

Summary

Hi, I'm Hemanth—an accomplished Software Engineer with 7+ years of experience building and scaling data-intensive, cloud-native applications. I specialize in designing resilient microservices and distributed data pipelines that process millions of records daily, while delivering seamless user experiences with React and TypeScript.

I hold an M.S. in Information Systems (Data Analytics) from NJIT (GPA 3.7).

My core technical toolkit spans:

-> Programming: Java 17 · Python · R · React · TypeScript · Bash Scripting
-> Frameworks: Spring Boot · Spring WebFlux · PySpark · Pandas · scikit-learn · Keras · R Shiny
-> Database: SQL · NoSQL · PostgreSQL · MySQL · MongoDB · Snowflake
-> Big Data & ETL: Spark · Sqoop · NiFi · HDFS · IICS · Control-M · Autosys
-> Cloud: AWS (S3 · EMR · Lambda · Step Functions · SQS · CloudFormation · CloudWatch · DynamoDB)
-> Infra: Docker · Kubernetes · OpenShift
-> Frontend (20%): React · TypeScript · Tailwind CSS · Bootstrap · HTML5 · CSS3
-> DevOps: Jenkins · Spinnaker · Control-M · IICS · Apigee · HashiCorp Vault · Kiali
-> Analytics & BI: Tableau · Apache Superset · Power BI
-> Methodologies: Agile/Scrum · BPMN · Requirement Analysis
-> IDEs: IntelliJ · Eclipse · STS · Jupyter · Spyder · RStudio

In my current role at Freddie Mac, I've led the development of event-driven ingestion platforms and real-time data APIs using Java Spring WebFlux, MongoDB, and AWS. I've also built React-based uploader tools that reduced errors by 75%, automated ZIP payload ingestion from third-party APIs using Bash, and transformed datasets with PySpark for analytics in Snowflake.

Prior to that, at LabCorp, I engineered backend services and caching layers in Spring Boot with Redis, built R Shiny dashboards on Cloudera, and automated patient data ingestion via Apache NiFi. At VentureSpace, I deployed FastAPI services, optimized SQL queries, and created Superset/Tableau dashboards that helped increase conversions by 20%.

I'm passionate about building scalable systems, reducing ETL latency, and designing intuitive platforms that turn raw data into actionable insight.

Let's connect to discuss scalable architectures, data-driven innovation, or opportunities to collaborate on cutting-edge software solutions!

Experience

Freddie Mac

Lead Software Engineer (Java, Spring WebFlux, NoSQL, PySpark, Bash, TypeScript, React, IIICS)

May 2020 - Present (6 years 5 months)

McLean, VA

- Led delivery of a web application built on Java 17, Spring WebFlux, and React + TypeScript for processing 2M+ appraisals/month on AWS EKS.
- Built Java backend application to process XML files in S3 via SQS poll and convert to JSON, enrich via external APIs and store in MongoDB.
- Designed File uploader component using React, TypeScript and styled with Tailwind CSS, cutting upload errors by 75% and latency by 30%.
- Delivered multiple GraphQL APIs over MongoDB, secured with Apigee, to deliver sub-second responses adopted by 27+ downstream teams.
- Built Agentic AI workflows using LangGraph, MCP, and tool calling to automate Jira story creation, task execution, code review, testing, and PR.

- Developed RAG and incident-triage agents with LangSmith tracing to answer project queries, summarize Jenkins failures, and create L3 incidents.
- Developed scalable PySpark jobs on AWS EMR to enrich 1M+ records per batch with scrubbed addresses and external metadata sources.
- Automated daily NAS-to-S3 file transfers using Control-M file watchers and Informatica CDI, completely removing manual processing steps.
- Implemented Bash scripts to retrieve ZIP files from Collibra and Workday APIs, extracted and processed the content, and uploaded to S3.
- Resolved all FOSS and SAST vulnerabilities across critical applications, reducing security findings by 100% to ensure audit compliance.
- Deployed interactive Kibana dashboards with real-time pod health and error log alerts to notify L3 and business teams during incidents.
- Authored detailed knowledge-base articles and conducted regular demos and handovers for support, operations, and business teams.

Labcorp

Software Engineer (Java, SpringBoot, R, HiveQL, Sqoop)

June 2019 - May 2020 (1 year)

Durham, NC

- Developed a robust Java Spring Boot application to retrieve patient and service center data from a DB2 database for healthcare reporting.
- Implemented caching using Spring Cache and Redis to reduce database calls and improve response time for frequently accessed endpoints.
- Built scalable R Shiny web app on Cloudera for self-service dashboards, increasing stakeholder satisfaction by 40% with real-time insights.
- Migrated legacy data from Oracle database to Hive using Sqoop and HiveQL, significantly reducing batch runtimes from 45 to 12 minutes.
- Automated Datameer workbook generation for genetics analytics, cutting report delivery from 3 days to under 2 hours for stakeholders.

VentureSpace

Software Engineer (Python, web scraping, FastAPI, SQL, Apache Superset, Tableau)

August 2018 - May 2019 (10 months)

New York, United States

- Built robust Python web scrapers aggregating listings from 10+ sites daily, tripling data coverage and reducing manual research time by 80%.
- Deployed scalable FastAPI microservice with Pydantic models and 95% test coverage, delivering live valuations under 50 ms for broker tools.

- Optimized complex PostgreSQL queries and materialized views, cutting dashboard load times from 12s to 1.5s for real-time trend analysis.
- Created 15+ interactive dashboards in Superset and Tableau, surfacing key business metrics that drove a 20% boost in listing conversions.

Education

New Jersey Institute of Technology

Master's degree, Information Systems, Data Analytics

Anna University

Bachelor's of Engineering , Electronics and Communications Engineering